

Lecture 5. Many particle states.

* Generalized Fourier transform

Consider

$$S(\mathbb{R}) := \left\{ \phi \in C^\infty(\mathbb{R}, \mathbb{C}) \text{ s.t. } \sup_{\alpha \in \mathbb{R}} \left| x^\alpha \frac{\partial \phi}{\partial x} \right| < \infty \quad \forall \alpha, \beta \geq 0 \right\}$$

(Schwartz space)

$$\rightarrow D(\mathbb{R}) \subset_{\text{dense}} S(\mathbb{R})$$

$\rightarrow S'(\mathbb{R})$: space of tempered distribution.

It is widely known that Fourier transform F ,

$F : S(\mathbb{R}) \rightarrow S(\mathbb{R})$ is an automorphism.

If $T \in S'(\mathbb{R})$, then we define $FT \in S'(\mathbb{R})$ as

$$(FT)(\phi) := T(F\phi) \quad \forall \phi \in S(\mathbb{R})$$

$\rightarrow$ Fourier transform of (Tempered distribution) $\psi(x) = e^{ip_0 x}$ is:

$$(F\psi)(\phi) = \psi(F\phi) \quad \forall \phi \in S(\mathbb{R})$$

$$(:= \int_{\mathbb{R}} e^{ip_0 x} \int_{\mathbb{R}} \phi(y) e^{-ixy} dy dx)$$

$$\int_{\mathbb{R}} e^{ip_0 x} \int_{\mathbb{R}} \phi(y) e^{-ixy} dy dx \stackrel{\text{DCT}}{=} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} e^{ip_0 x} \cdot e^{-\epsilon x^2} \int_{\mathbb{R}} \phi(y) e^{-ixy} dy dx$$

$$= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \underbrace{\phi(y) e^{ix(p_0 - y) - \epsilon x^2}}_{\text{abs integrable}} dy dx$$

Fubini

$$= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(y) e^{ix(p_0 - y) - \epsilon x^2} dx dy$$

$$= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} \phi(y) \sqrt{\frac{\pi}{\epsilon}} e^{-\frac{(p_0 - y)^2}{4\epsilon}} dy$$

$$= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} \phi(2\sqrt{\epsilon}z + p_0) 2\sqrt{\pi} e^{-z^2} dz = 2\pi \phi(p_0)$$

$$\rightarrow (F\psi)(\phi) = 2\pi \delta_0(\phi) \rightarrow \hat{\psi} = 2\pi \delta_p$$

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$$\begin{aligned} &= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}} \phi(2\sqrt{\epsilon}z + p_0) 2\sqrt{\pi} e^{-z^2} dz = 2\pi \phi(p_0) \\ &\rightarrow (F\psi)(\phi) = 2\pi \delta_0(\phi) \rightarrow \hat{\psi} = 2\pi \delta_0 \end{aligned}$$

$$H_x = -\frac{1}{2m} \frac{d^2}{dx^2}, \quad \psi_t = e^{-itH_x} \psi, \quad \varphi = \hat{\psi}, \quad \varphi_t = e^{-\frac{it}{2m} p^2} \varphi(p), \quad H_p = \frac{p^2}{2m}$$

$$\rightarrow \underbrace{-\frac{1}{2m} \frac{d^2}{dx^2} \psi}_{\hat{\psi}'} = \frac{p^2}{2m} \hat{\psi} \quad (\text{Basic property of Fourier transform})$$

$$\hat{\psi}' = ip \hat{\psi}$$

5.2. Schrödinger & Heisenberg pictures.

state $\psi_t = e^{-itH} \psi, \quad \psi_t = \psi$

observable operator $A_t = A, \quad A_t = e^{itH} A e^{-itH}$

expected value $\langle \psi | e^{itH} A e^{-itH} | \psi \rangle \xrightarrow{\text{equivalence}} \langle \psi | e^{itH} A e^{-itH} | \psi \rangle$

• Tensor product state spaces

$$H^{\otimes n} := \overline{\text{span} \{ \psi_1 \otimes \dots \otimes \psi_n, \psi_i \in H \}}$$

Where: $\psi_1 \otimes \dots \otimes (\alpha \psi_i) \otimes \dots \otimes \psi_n = \alpha (\psi_1 \otimes \dots \otimes \psi_n)$ (scalar multiples)

$$\psi_1 \otimes \dots \otimes (\psi_i + \psi_i') \otimes \dots \otimes \psi_n = (\psi_1 \otimes \dots \otimes \psi_i) \otimes \dots \otimes \psi_n + (\psi_1 \otimes \dots \otimes \psi_i') \otimes \dots \otimes \psi_n$$

(distributivity)

$$(\psi_1 \otimes \dots \otimes \psi_n, \phi_1 \otimes \dots \otimes \phi_n) := (\psi_1, \phi_1) \dots (\psi_n, \phi_n)$$

If $H = L^2(\mathbb{R})$ then $H^{\otimes n} \cong L^2(\mathbb{R}^n)$

$$(\psi_1 \otimes \dots \otimes \psi_n)(x_1, \dots, x_n) = \psi_1(x_1) \cdot \dots \cdot \psi_n(x_n) \quad \forall \psi_i \in L^2(\mathbb{R}), x_i \in \mathbb{R}$$

Lecture 6. Bosonic Fock space.

$\therefore$ n identical n particles are indistinguishable.

Let $\hat{P}_{ij}$: exchange operator s.t.

$$\hat{P}_n \mathbb{E}(x_1, \dots, x_{i-1}, x_{i+1}, x_n) = \mathbb{E}(x_1, \dots, x_{i-1}, x_{i+1}, x_n)$$

Clearly, $\hat{p}_\mu$ is self-adjoint and $\hat{p}_\mu^2 = \mathbb{I}_d$.

→ eigen values are either 1 or -1

→ eigenspaces E_{+i} E_{-i}

We set $\mathcal{H}_{\text{Sym}}^{\otimes n} := \bigcap_{i \neq j} E_{+}^{ij}$, $\mathcal{H}_{\text{Alt}}^{\otimes n} := \bigcap_{i \neq j} E_{-}^{ij}$

$$\rightarrow \mathcal{H}_{Sym}^{\otimes n} = \overline{\text{span} \left\{ \sum_{\sigma \in S_n} \psi_{\sigma(1)} \otimes \dots \otimes \psi_{\sigma(n)} : \psi_1, \dots, \psi_n \in \mathcal{H} \right\}}$$

Main postulate: state of n identical Bosons $\in \mathcal{H}_{\text{sym}}^{\otimes n}$

1. Fermions $\in \mathcal{H}_{Alt}^{\otimes n}$.

↳. If $\gamma_i = \gamma_j$ for some $i \neq j$, then

$$\sum_{\sigma \in S_n} (-1)^{|\sigma|} \psi_{\sigma(1)} \otimes \cdots \otimes \psi_{\sigma(n)} = 0.$$

(i.e. Pauli exclusion principle)

• Time evolution on $H^{\otimes n}$.

If $(U(t))_{t \in \mathbb{R}}$ is a unitary group, then

$$U(t)(\psi_1 \otimes \dots \otimes \psi_n) := (U(t)\psi_1) \otimes \dots \otimes (U(t)\psi_n)$$

The corresponding self-adjoint operator $A = -iH$ is defined as

$$A\psi = \lim_{t \rightarrow 0} \frac{U(t)\psi - \psi}{it} \quad \text{by Stone's thm.}$$

$$\begin{aligned} \therefore H(\psi_1 \otimes \dots \otimes \psi_n) &= -i \lim_{t \rightarrow 0} \frac{1}{it} \left(U(t)(\psi_1 \otimes \dots \otimes \psi_n) - \psi_1 \otimes \dots \otimes \psi_n \right) \\ &= -i \lim_{t \rightarrow 0} \frac{1}{it} \sum_{j=1}^n (U(t)\psi_j) \otimes \dots \otimes (U(t)\psi_{j-1}) \otimes (U(t)\psi_{j+1} - \psi_{j+1}) \otimes \dots \otimes \psi_n \\ &= \sum_{j=1}^n \psi_1 \otimes \dots \otimes \psi_{j-1} \otimes H\psi_j \otimes \psi_{j+1} \otimes \dots \otimes \psi_n \end{aligned}$$

• In particular, if $\mathcal{H} = L^2(\mathbb{R})$, $H = -\frac{1}{2m} \frac{d^2}{dx^2}$.

$$\rightarrow \mathcal{H}^{\otimes n} = L^2(\mathbb{R}^n),$$

$$H(\psi_1 \otimes \dots \otimes \psi_n) = \sum_{j=1}^n \psi_1 \otimes \dots \otimes \psi_{j-1} \otimes \left(-\frac{1}{2m} \psi_j'' \right) \otimes \psi_{j+1} \otimes \dots \otimes \psi_n.$$

$$\begin{aligned} H(\psi_1 \otimes \dots \otimes \psi_n)(x_1, \dots, x_n) &= -\frac{1}{2m} \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2} (\psi_1(x_1) \dots \psi_n(x_n)) \\ &= -\frac{1}{2m} \Delta(\psi_1 \otimes \dots \otimes \psi_n) \end{aligned}$$

$$\rightarrow H\psi = -\frac{1}{2m} \Delta\psi \quad \forall \psi \in D(\Delta) : \text{domain of the unique self-adj. extension of } \Delta.$$

• Bosonic Fock space

$$\mathcal{B} := \{(\psi_0, \psi_1, \dots) : \psi_n \in \mathcal{H}_{sym}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty\}$$

inner product of $\psi = (\psi_0, \psi_1, \dots)$ and $\phi = (\phi_0, \phi_1, \dots)$ is

$$(\psi, \phi) = \sum_{n=0}^{\infty} (\psi_n, \phi_n) \rightarrow \text{Hilbert space}$$

system has n particles with prob $\frac{\|\psi_n\|^2}{\|\psi\|^2}$

$$\mathcal{B}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}_{sym}^{\otimes n} \subset \mathcal{B}$$

dense.

we can do similar work to Fermion:

$$\mathcal{F} := \{(\psi_0, \psi_1, \dots) : \psi_n \in \mathcal{H}_{alt}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty\}$$

inner product also defined as $(\psi, \phi) = \sum_{n=0}^{\infty} (\psi_n, \phi_n)$

$$\mathcal{F}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}_{alt}^{\otimes n} \subset \mathcal{F}$$

dense.

This process: constructing Fock space,

counting the number of particles in a given state in occupation number basis $|n_1, n_2, \dots\rangle$

is called as second quantization.

We can also use $|n_1, n_2, \dots\rangle$ to Fermion, where

$$|n_1, n_2, \dots\rangle = (\text{normalization of}) \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}} \times (-1)^{|\sigma|}$$

where n_i : number of i 's in $\{m_1, \dots, m_n\}$

$\rightarrow n_i = 0$ or $1, \forall i$. (Pauli's principle)

If $\{e_1, e_2, \dots\}$ is an ONB of $\mathcal{H}$, then

$\left\{ \sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}}, m_1, \dots, m_n \in \mathbb{N} \right\}$ is an orthogonal basis of $\mathcal{H}_{\text{sym}}^{\otimes n}$

inner product with $\sum_{\sigma \in S(n)} e_{m'_{\sigma(1)}} \otimes \dots \otimes e_{m'_{\sigma(n)}}$ is nonzero only if

$$\{m_1, \dots, m_n\} = \{m'_1, \dots, m'_n\}$$

i.e.

$$n_i = |\{j: m_j = i, 1 \leq j \leq n\}|$$

$$n'_i = |\{j: m'_j = i, 1 \leq j \leq n\}|$$

are equal for every i .

$$\left\| \sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}} \right\|^2$$

$$= \sum_{\sigma, \tau \in S(n)} \delta_{m_{\sigma(1)} m_{\tau(1)}} \dots \delta_{m_{\sigma(n)} m_{\tau(n)}}$$

(For each $\sigma \in S(n)$, we can find τ satisfying $m_{\sigma(i)} = m_{\tau(i)} \quad \forall i \in \{1, \dots, n\}$)

$$= n! \prod_{i=1}^{\infty} n_i!$$

$\rightarrow \left\{ \frac{1}{\sqrt{n! \prod_{i=1}^{\infty} n_i!}} \sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}}, m_1, \dots, m_n \in \mathbb{N} \right\}$ is an ONB of $\mathcal{H}_{\text{sym}}^{\otimes n}$

Since $\left(\sum_{\sigma \in S(n)} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}} \right)$ can be identified by $\{m_1, \dots, m_n\}$

we can denote it by $|n_1, n_2, \dots\rangle$.

For simplicity, we also define $\mathcal{H}_{\text{sym}}^{\otimes 0}$ as $\mathbb{C}$, with $\mathbb{C}$ -basis $\{1\}$

$\rightarrow$ we denote $| := |0, 0, 0, \dots\rangle$ (This is called the vacuum state)

Similarly,

$$a_k \left(\frac{1}{\sqrt{n!} \prod_{i=1}^n n_i!} \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \right)$$

$$= \frac{1}{\sqrt{(n-1)! (n_1-1)! \cdots n_k!}} \sum_{\tau \in S_{n-1}} e_{m'_{\tau(1)}} \otimes \cdots \otimes e_{m'_{\tau(n-1)}}$$

$$\rightarrow a_k \left(\sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}} \right)$$

$$= \sqrt{n} n_k \sum_{\tau \in S_{n-1}} e_{m'_{\tau(1)}} \otimes \cdots \otimes e_{m'_{\tau(n-1)}} \quad \text{--- (xx)}$$

where $\{m'_1, \dots, m'_{n-1}\} = \{m_1, \dots, m_n\} - \{k\}$.

On the other hand,

$$\sum_{\sigma \in S_n} (e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n)}}) \times e_k^*(e_{m_{\sigma(n)}})$$

$$= \sum_{\substack{\sigma \in S_n \\ m_{\sigma(n)} = k}} e_{m_{\sigma(1)}} \otimes \cdots \otimes e_{m_{\sigma(n-1)}}$$

$$= n_k \sum_{\tau \in S_{n-1}} e_{m'_{\tau(1)}} \otimes \cdots \otimes e_{m'_{\tau(n-1)}}$$

$$\rightarrow \text{A linear operator } a_k: \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \rightarrow \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \quad \text{s.t.}$$

$$(a_k)(\psi_1 \otimes \cdots \otimes \psi_n) = \sqrt{n} (\psi_1 \otimes \cdots \otimes \psi_{n-1}) \times (e_k^*)(\psi_n)$$

satisfies (xx)

$\rightarrow \{e_1, e_2, \dots\}$ corresponds to $\{a_1, a_2, \dots\}$ by this way.

$$\xrightarrow{\text{we extend}} A^*(f) (\psi_1 \otimes \cdots \otimes \psi_n) = \sqrt{n} (\psi_1 \otimes \cdots \otimes \psi_{n-1}) \times \langle f, \psi_n \rangle$$

(since $\psi_n f \in \mathcal{H}$, $f^*(\psi_n) := \langle f, \psi_n \rangle$)

Lecture 7. Creation and annihilation operators.

For given ONB $\{e_1, e_2, \dots\}$ of $\mathcal{H}$, the associated

$$\text{creation} : a_k^+ |n_1, n_2, \dots\rangle = \sqrt{n_k+1} |n_1, \dots, n_k+1, n_{k+1}, n_{k+2}, \dots\rangle.$$

$$\text{i.e. } a_k^+ : \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \rightarrow \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \text{ satisfy}$$

$$a_k^+ \cdot \left(\frac{1}{\sqrt{n!} \prod_{i=1}^{\infty} n_i!} \sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}} \right) \\ = \frac{1}{\sqrt{(n+1)!} \prod_{i=1}^{\infty} n_i!} \sum_{\tau \in S_{n+1}} e_{m'_{\tau(1)}} \otimes \dots \otimes e_{m'_{\tau(n+1)}} \quad (*)$$

$$\text{where } m'_1 = m_1, \dots, m'_n = m_n, m'_{n+1} = k.$$

$$\rightarrow a_k^+ \left(\sum_{\sigma \in S_n} e_{m_{\sigma(1)}} \otimes \dots \otimes e_{m_{\sigma(n)}} \right) = \frac{1}{\sqrt{n+1}} \sum_{\tau \in S_{n+1}} e_{m'_{\tau(1)}} \otimes \dots \otimes e_{m'_{\tau(n+1)}}$$

Note that for each $i=1, \dots, n+1$,

$$\{\tau \in S_{n+1} : \tau(i) = n+1\} \text{ is isomorphic to } S_n.$$

$$\rightarrow \text{A linear operator } a_k^+ : \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \rightarrow \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \text{ s.t.}$$

$$a_k^+ (\psi_1 \otimes \dots \otimes \psi_n) = \frac{1}{\sqrt{n+1}} \left(\begin{aligned} &\psi_1 \otimes \dots \otimes \psi_n \otimes (e_k) \\ &+ \psi_1 \otimes \dots \otimes \psi_{n-1} \otimes (e_k \otimes \psi_n) \\ &+ \dots \\ &+ (e_k \otimes \psi_1 \otimes \dots \otimes \psi_n) \end{aligned} \right).$$

satisfy (*). So $\{e_k\}_{k \in \mathbb{Z}_+}$ corresponds to $\{a_k^+\}_{k \in \mathbb{Z}_+}$ by this.

$$\rightarrow A^+(f) (\psi_1 \otimes \dots \otimes \psi_n) := \frac{1}{\sqrt{n+1}} \left(\begin{aligned} &\psi_1 \otimes \dots \otimes \psi_n \otimes f \\ &+ \dots \\ &+ f \otimes \psi_1 \otimes \dots \otimes \psi_n \end{aligned} \right).$$

We extend

If $\mathcal{H} = L^2(\mathbb{R})$, then

$$(A(f)\varphi)(x_1, \dots, x_{n+1}) = \sqrt{n} \int_{-\infty}^{\infty} \overline{f(y)} \varphi(x_1, \dots, x_n, y) dy$$

$$(A^+(f)\varphi)(x_1, \dots, x_{n+1}) = \frac{1}{\sqrt{n+1}} \sum_{j=1}^{n+1} f(x_j) \varphi(x_1, \dots, \hat{x}_j, \dots, x_{n+1})$$

$\forall \varphi \in L^2(\mathbb{R}^n)_{\text{sym}}$

$\forall \varphi \in L^2(\mathbb{R}^n)_{\text{sym}}$

$$a_k = A(e_k) \quad a_k^+ = A^+(e_k)$$

$$\rightarrow a(x) = A(\delta_x); \quad a^+(x) = A^+(\delta_x)$$

Heaviside

$$(a(x)\varphi)(x_1, \dots, x_{n+1}) = \sqrt{n} \varphi(x_1, \dots, x_n, x)$$

$$(a^+(x)\varphi)(x_1, \dots, x_{n+1}) = \frac{1}{\sqrt{n+1}} \sum_{j=1}^{n+1} \delta(x-x_j) \varphi(x_1, \dots, \hat{x}_j, \dots, x_{n+1})$$

$$\rightarrow A(f) = \int dx \overline{f(x)} a(x), \quad A^+(f) = \int dx f(x) a^+(x)$$

$$a_k a_l^+ - a_l^+ a_k = 0 \quad \text{if } k \neq l, \quad \text{since } |n_1, n_2, \dots\rangle \rightarrow \sqrt{n_k+1} \sqrt{n_k} | \dots n_k-1, \dots n_{k+1} \rangle$$

$$1 \quad \text{if } k=l, \quad \text{since } a_k a_k^+ = (n_k+1)1, \quad a_k^+ a_k = n_k 1.$$

On the other hand, since

$$[A(\sum_k \alpha_k e_k), A^+(\sum_l \beta_l e_l)]$$

$$= \left[\sum_{k=1}^{\infty} \alpha_k^+ a_k, \sum_{l=1}^{\infty} \beta_l a_l^+ \right]$$

$$= \sum_{k,l} \alpha_k^+ \beta_l [a_k, a_l^+]$$

$$= \sum_{k,l} \alpha_k^+ \beta_l \delta_{k,l} 1 = (\sum_k \alpha_k e_k, \sum_l \beta_l e_l) 1$$

$$\text{and } [A(f), A^+(g)] = \int dx dy \overline{f(x)} g(y) [a(x), a^+(y)],$$

$$\text{we have } [a(x), a^+(y)] = \delta(x-y) 1.$$

Similarly, since $[a_k, a_k] = [a_k^+, a_k^+] = 0$, we have

$$[a(x), a(y)] = [a^+(x), a^+(y)] = 0, \quad \forall x, y \in \mathbb{R}.$$

Lecture 8 Time evolution on Fock space

If A is a linear operator on $L^2(\mathbb{R})$, $\varphi \in L^2(\mathbb{R}^n)_{\text{sym}}$, then

$$\left(\int dx a^\dagger(x) A a(x) \right) (\varphi)(x_1, \dots, x_n)$$

$$(a(x) \varphi)(x_1, \dots, x_{n-1}) = \sqrt{n} \varphi(x_1, \dots, x_{n-1}, x)$$

$$(A_x a(x) \varphi)(x_1, \dots, x_n) = A_x (\sqrt{n} \varphi(x_1, \dots, x_n, x))$$

$$\begin{aligned} (a^\dagger(x) A_x a(x) \varphi)(x_1, \dots, x_n) &= \sum_{j=1}^n \delta(x - x_j) A_x (\varphi(x_1, \dots, x_n, x)) \\ &= \sum_{j=1}^n A_{x_j} (\varphi(x_1, \dots, x_n)) \end{aligned}$$

If A is multiplication operator f , then

$$\begin{aligned} \left(\int dx a^\dagger(x) A a(x) \right) \varphi &= \int dx f(x) a^\dagger(x) a(x) \varphi = \sum_{j=1}^n A_{x_j} \varphi(x_1, \dots, x_n) \\ &= \left(\sum_{j=1}^n f(x_j) \right) \varphi(x_1, \dots, x_n) \end{aligned}$$

If A is free evolution Hamiltonian $H = -\frac{1}{2m} \frac{d^2}{dx^2}$

$$\begin{aligned} \therefore \left(\int_{\mathbb{R}} dx a^\dagger(x) A a(x) \right) \varphi &= \sum_{j=1}^n A_{x_j} \varphi(x_1, \dots, x_n) \\ &= -\frac{1}{2m} \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2} \varphi(x_1, \dots, x_n) = -\frac{1}{2m} \Delta \varphi \end{aligned}$$

If $\varphi = \sum_{i \in \mathbb{N}} \varphi_i \otimes \dots \otimes \varphi_n$, then

$$\int dx a^\dagger(x) A a(x) \varphi = \sum_{i=1}^n (A_{x_i} (\varphi(x_1, \dots, x_n))) = \sum_{i=1}^n \varphi_i(x_i) \dots \varphi_{i-1}(x_{i-1}) (A \varphi_i(x_i)) \dots \varphi_n(x_n)$$

This is exactly how Hamiltonian for tensor product modes (Lecture 54)

I think there are nothing new in 8.4 and 8.5, so let's finish today's seminar.